Jinqiu Du (Turbo)

turbodu@uw.edu | +1 (206) 370-4374 | https://turbodu222.github.io/

EDUCATION

University of Wisconsin - Madison Sep. 2026 – Jun. 2031
Ph.D. in Biomedical Data Science Madison, WI

University of Washington Sep. 2024 – Mar. 2026
M.S. in Biostatistics – Modeling and Methods Track Seattle, WA

Hong Kong Baptist University Sep. 2020 – Jun. 2024
B.S. (Honors) in Statistics (First Class) Hong Kong

PUBLICATIONS & PRESENTATIONS

[1] **J. Du** and K. Z. Lin, "Integrating morphology and gene expression of neural cells in unpaired single-cell data using GeoAdvAE," Accepted at *Research in Computational Molecular Biology (RECOMB),* 2026*; bioRxiv (preprint), 2025*.
- Presented at the *American Society of Human Genetics (ASHG) Annual Meeting*, 2025.

[2] W. Zhang, **J. Du**, Y. Chen, S. Jayadev, K. E. Prater, and K. Z. Lin, "Large-scale snRNA-seq meta-analysis of microglia role in Alzheimer's disease across statistical methods," preprint, 2025.

[3] J. Gong, **J. Du**, and S. R. Hingorani, "Isolated nocturnal hypertension is associated with the development of daytime hypertension in otherwise healthy children," accepted for presentation at *the Western Medical Research Conference*, 2026.

[4] H. Bender, **J. Du**, and S. R. Hingorani, "The association of puberty and compensatory hypertrophy on kidney function in pediatric patients with congenital solitary functioning kidney," accepted for presentation at the *Western Medical Research Conference*, and accepted for poster presentation at the *Pediatric Academic Societies (PAS) Conference*, 2026.

[5] S. Liu, **J. Du**, Y. Zheng, J. Yin, Y. Deng, and J. Wu, "Energy-Aware Holistic Optimization in UAV-Assisted Fog Computing: Attitude, Trajectory, and Task Assignment," *Computer Networks, 277:112064,* 2026.

[6] S. Liu, **J. Du**, Y. Zheng, J. Yin, Y. Deng, and J. Wu, "Holistic optimization in UAV-assisted fog computing: attitude, trajectory, task assignment," under minor revision for *Computer Networks*, arXiv (preprint), 2024.

[7] S. Liu, J. Yin, **J. Du,** Y. Zheng, Y. Deng, and J. Wu, "Meteorological and topographical big data-driven UAV trajectory planning," *Proceedings of the 34th IEEE International Telecommunication Networks and Applications Conference*, 2024.

[8] **J. Du**, Y. Zheng, S. Liu, J. Luo, J. Yin, Y. Deng, and J. Wu, "Optimizing vaccine and ventilator allocation to minimize health risks and costs during pandemics," under review at *Health Care Management Science*, preprint, 2024.

RESEARCH EXPERIENCE

Integrating Neural Cell Morphology and Gene Expression for Cross-Modal Analysis Jan. 2025 – Present
First author (Supervised by Prof. Kevin Z. Lin, University of Washington)

- Led the development of GeoAdvAE, the first model to link neural cell's physical shape with its gene expression even across unpaired datasets, achieving ~20% higher alignment accuracy than 11 existing methods on Patch-seq and simulated data.
- Developed interpretable simulations that generated realistic neuron shapes, allowing us to test whether the model could recover biologically meaningful morphology–gene relationships.
- Applied GeoAdvAE to unpaired microglia data from 5xFAD Alzheimer's model mice, revealing morphology-associated genes that distinguish diverse microglial states relevant to Alzheimer's pathology.

* Code and documentation are publicly available at https://github.com/turbodu222/GeoAdVAE

Spatial Transcriptomics-Guided Deep Learning for Lung Cancer Pathology Image Analysis Feb. 2025 – Present
Research Assistant (Supervised by Prof. Wei Sun, Fred Hutchinson Cancer Center)

- Developed a multimodal analysis pipeline integrating single-cell RNA-seq, spatial transcriptomics, and histopathology image features to identify spatial cell-type distributions in lung cancer.
- Proposed a modified CARD R package enabling custom differentially expressed genes (DEGs) for cell-type deconvolution.
- Extracted latent representations from multiple vision foundation models (CONCH, ProvGigapath, Virchow, UNI2h, etc.) to learn high-dimensional feature embeddings from histopathology slide images.
- Trained an XGBoost model in Python using histology-derived embeddings to predict cell-type composition, achieving a Pearson correlation of ~0.7 between predicted and true proportions on held-out test samples.

* Manuscript based on this work is currently in preparation.

Cohort-Level Differential Distributional Analysis of Microglia in Alzheimer's Disease via Single-Nuclei RNA-Seq
Co-author (Supervised by Prof. Kevin Z. Lin & UW Neuroscience) Sep. 2024 – Jun. 2025

- Studied the reproducibility of microglial transcriptomic differences among people with and without Alzheimer's disease based on single-cell RNA-seq data across multiple cohorts (including Prater, SEA-AD, and ROS-MAP datasets).
- Performed extensive down-sampling experiments (artificially removing donors, cells, or sequences) to assess how the performance of various differential-expression methods deteriorates as data was removed.
- Benchmarked multiple differentially expressed (DE) methods (edgeR, DESeq2, eSVD, and an SCVI-based latent variable DE framework) to find pathways that were reproducible across different methods.

Survival Modeling of Abnormal Ambulatory Blood Pressure and Kidney Function Jul. 2025 – Sep. 2025
Research Assistant (Collaborated with Prof. Sangeeta R. Hingorani, Seattle Children's Hospital)

- Proposed an adaptive Cox modeling framework with time-dependent covariate updating to address irregular follow-up and informative censoring, improving dynamic risk estimation of daytime hypertension in pediatric cohorts.
- Developed a novel REML-imputation hybrid mixed model to jointly address missing at random (MAR) data and within-subject correlation, enabling robust estimation of complex nonlinear eGFR trajectories over time.

* Received the Best Gastroenterology, Hepatology & Nephrology Abstract Award from Western Society for Pediatric Research.

Optimization of Healthcare Resource Allocation During the COVID-19 Pandemic Jan. 2023 – Dec. 2024
First author (Supervised by Prof. Timothy Jingjin Wu & Prof. Yuhui Deng)

- Developed a novel optimization framework integrating an extended SVUIR epidemiological model with a linearized mixed-integer programming (MIP) formulation to jointly allocate vaccines and ventilators under dynamic outbreak conditions.
- Applied the framework to real COVID-19 data across 17 age groups and 5 regions over a 400-day horizon, achieving >60% higher computational efficiency and up to 70% cost reduction compared with baseline strategies.

Optimization of UAV Trajectory Planning and Task Allocation in Fog Computing Jun. 2022 – Dec. 2024
Project Director (Supervised by Prof. Timothy Jingjin Wu & Prof. Yuhui Deng)

- Developed an adaptive control algorithm to ensure UAV flight stability under dynamic weather and terrain conditions.
- Designed an enhanced ant colony optimization framework for 3D trajectory planning and task assignment, tested on real weather/terrain data, improving crash prediction and boosting energy efficiency by >34% with 2.6× higher task completion.

* Independently funded by the Provincial Government's "*Climbing Plan*" research grant.

TEACHING & INTERNSHIP EXPERIENCE

Teaching Assistant, University of Washington
BIOST 546: *Machine Learning for Biomedical and Public Health Big Data (Graduate course)* Dec. 2025 – Present

- Graded homework, prepared keys, and helped students with Python and R programming during office hours.

BIOST 310: *Biostatistics for the Health Sciences (Undergraduate course)* Sep. 2025 – Dec. 2025

- Led 2-hour discussion sessions per week, held office hours, and graded coursework for ~200 students. Seattle, WA

Tutor in Statistics, University of Washington Sep. 2024 – Present
Statistics Study Center, Department of Statistics Seattle, WA

- Assisted ~30 undergraduate students weekly in statistical coursework and R programming during 4-hour tutoring sessions.

Volunteer Contributor, University of Washington May 2025 – Aug. 2025
BIOST 545: *Biostatistical Methods for Big Omics Data (Graduate course)* Seattle, WA

- Built the course website with R (Quarto framework) to present lecture materials and reproducible code examples.

National Bureau of Statistics of China Jun. 2023 – Sep. 2023
Modeling Assistant, Agricultural Survey Division Chengdu, CHN

- Built regression models from remote sensing data to estimate rural land use and visualized output for a 50-page official atlas.

HONORS & AWARDS

Finalist, *COMAP Mathematical Contest in Modeling (MCM),* Top 1% Worldwide 2022
First-Class Academic Scholarship, *Hong Kong Baptist University,* Top 5% 2021–2023
President's Honour Roll, *Hong Kong Baptist University* 2020–2024
Meritorious Winner, *COMAP Mathematical Contest in Modeling (MCM)*, Top 6% Worldwide 2023
Student Internship Scholarship, *Hong Kong Baptist University* 2021

PROGRAMMING SKILLS

Python, R, Linux, C, MATLAB, LaTeX, Git, SQL, SPSS, Lingo, Excel, Affinity Designer